

## Project Design Phase-II Technology Stack (Architecture & Stack)

|               |                                              |
|---------------|----------------------------------------------|
| Date          | 13 March 2026                                |
| Team ID       | 69bf7fca443e4a653d45f826                     |
| Project Name  | Smart Library Request Workflow in ServiceNow |
| Maximum Marks | 4 Marks                                      |

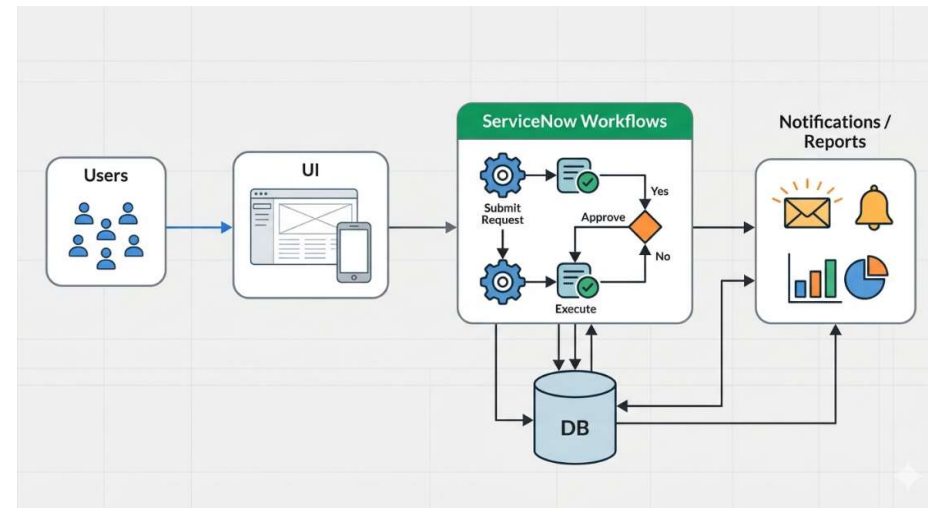
### Technical Architecture

The Smart Library Request Workflow is designed as a **cloud-based service management application** using **ServiceNow** as the main platform. It integrates user-facing interfaces with backend workflows, automated approvals, and database storage for tracking requests, issuance, and reporting

### Key Features of the Architecture:

- **User Layer:** Web-based interface accessible via desktop and mobile browsers.
- **Application Layer:** ServiceNow platform handles workflows, approvals, notifications, and business logic.
- **Integration Layer:** APIs for email notifications and authentication (Gmail, LinkedIn).
- **Database Layer:** ServiceNow's cloud database for storing user and request data.
- **Reporting Layer:** Dashboard and analytics for library admin.

Reference: [IBM Pandemic Order Processing Example](#)



**Table-1: Components & Technologies**

| S.No | Component           | Description                                    | Technology / Tool Used                               |
|------|---------------------|------------------------------------------------|------------------------------------------------------|
| 1    | User Interface      | How users interact with the library workflow   | ServiceNow Web UI, HTML, CSS, JavaScript             |
| 2    | Application Logic-1 | Workflow automation for book requests          | ServiceNow Flow Designer, Business Rules             |
| 3    | Application Logic-2 | Approval process for librarian                 | ServiceNow Approval Engine                           |
| 4    | Application Logic-3 | Notifications and email triggers               | ServiceNow Notification Engine, Gmail API            |
| 5    | Database            | Storage for user profiles, requests, book info | ServiceNow Cloud Database (MySQL variant)            |
| 6    | Cloud Database      | ServiceNow backend cloud DB for scalability    | ServiceNow Cloud Database                            |
| 7    | File Storage        | Store attachments like request forms           | ServiceNow Attachment Storage                        |
| 8    | External API-1      | User authentication via Gmail                  | Google OAuth API                                     |
| 9    | External API-2      | User authentication via LinkedIn               | LinkedIn OAuth API                                   |
| 10   | Machine Learning    | Optional: Suggest books based on user history  | Python ML model / ServiceNow Predictive Intelligence |
| 11   | Infrastructure      | Application Deployment on cloud                | ServiceNow Cloud, Multi-region deployment            |

**Table-2: Application Characteristics**

| S.No | Characteristics          | Description                                                                             | Technology / Tool Used                                        |
|------|--------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------|
| 1    | Open-Source Frameworks   | Frontend customization using HTML, CSS, JavaScript frameworks                           | AngularJS / ReactJS                                           |
| 2    | Security Implementations | User authentication, role-based access, encrypted data, secure OAuth for Gmail/LinkedIn | SHA-256, HTTPS, OAuth2, ServiceNow ACLs, OWASP Best Practices |
| 3    | Scalable Architecture    | 3-tier architecture with cloud backend, microservices workflow-ready                    | ServiceNow Cloud, Microservices-ready Flow Designer           |

|   |              |                                                                                      |                                                        |
|---|--------------|--------------------------------------------------------------------------------------|--------------------------------------------------------|
| 4 | Availability | Distributed servers and cloud redundancy ensure high availability                    | Multi-region ServiceNow Cloud, Load Balancers          |
| 5 | Performance  | Efficient request handling, caching, notifications optimized, minimal page load time | ServiceNow Cache, CDN for attachments, Async workflows |

#### References:

1. [C4 Model for Software Architecture](#)
2. [IBM Online Order Processing System during Pandemic](#)
3. [IBM Cloud Architecture](#)
4. [AWS Cloud Architecture](#)
5. [Technical Architecture Diagrams Guide](#)